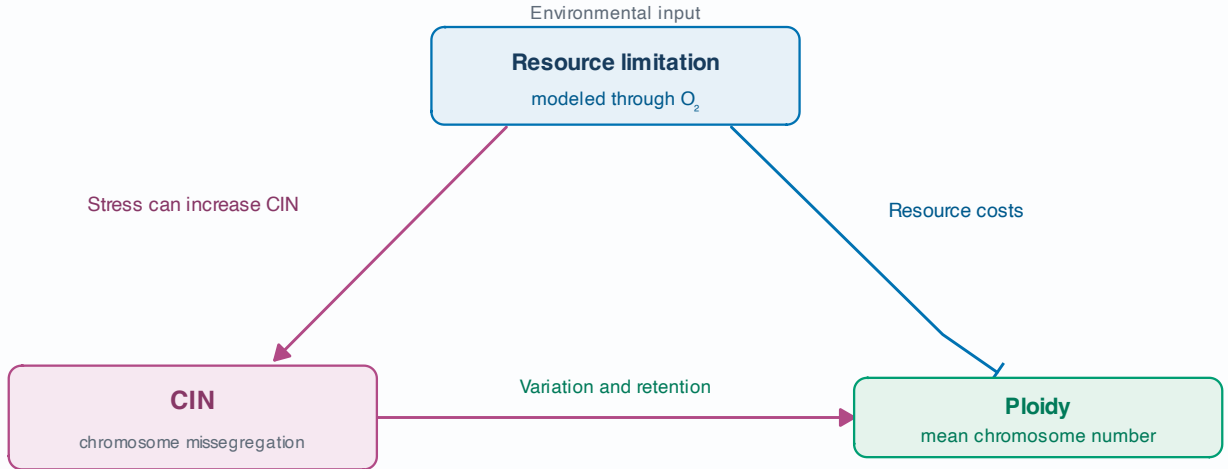
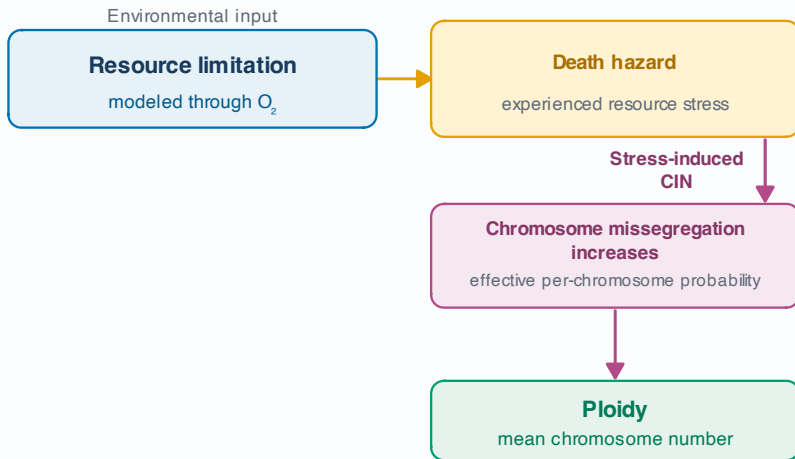


## Resource limitation pushes ploidy in two opposing directions



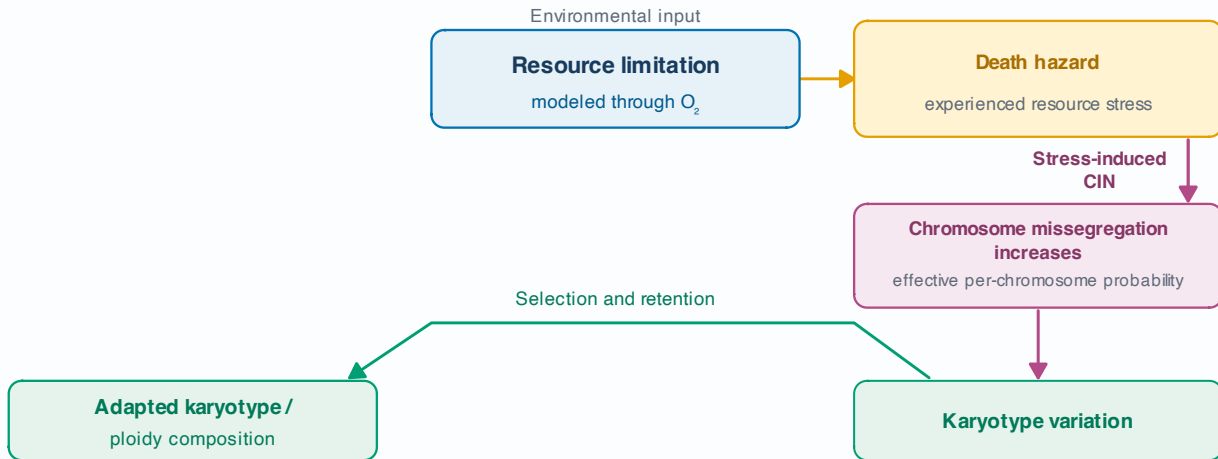
Resource costs can oppose high ploidy; buffering can favor its retention under CIN.

## Experienced resource stress links the environment to CIN



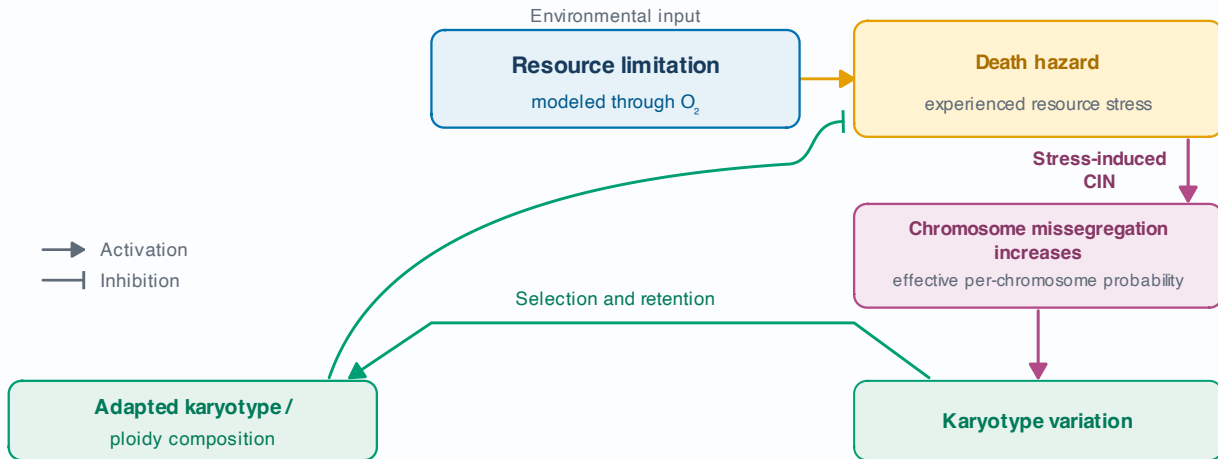
CIN responds to the modeled death hazard, not to oxygen alone.

## Generating variation and retaining adapted descendants are different processes



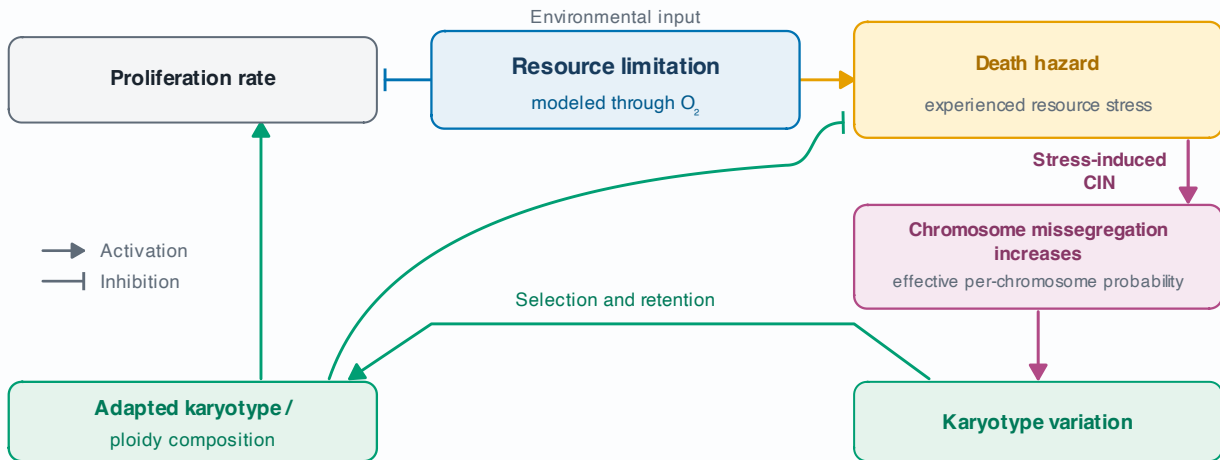
CIN supplies karyotype variation; selection and retention reshape the population.

## Stress-induced mutagenesis: adaptation can turn CIN back down



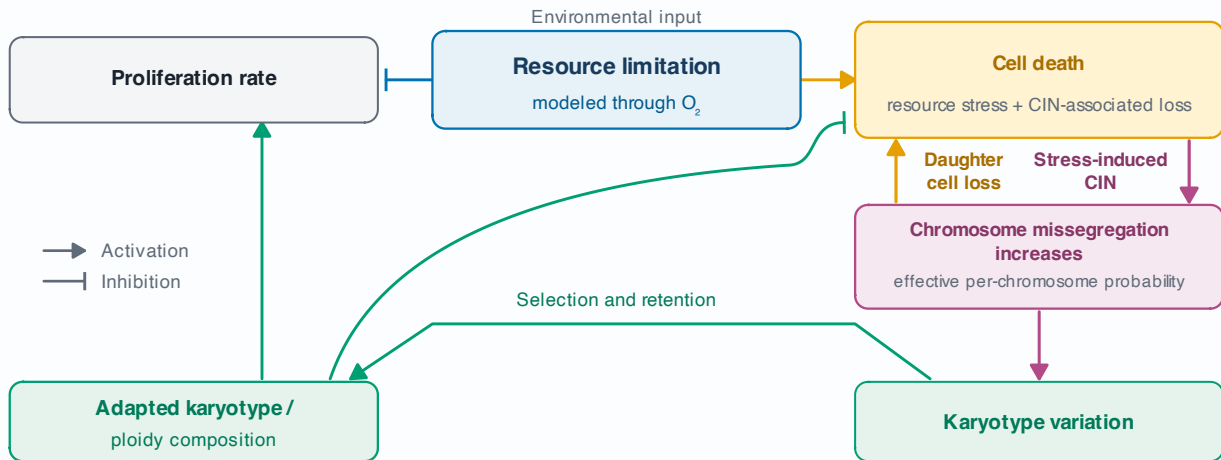
At unchanged oxygen, adaptation can reduce experienced stress and population-average CIN.

## Adapted composition also changes population growth



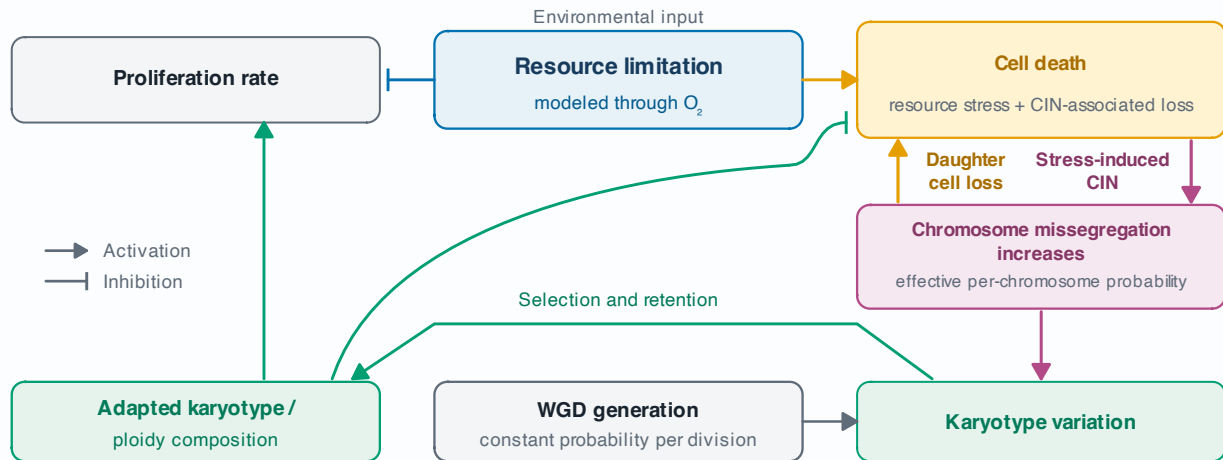
Resource limitation suppresses proliferation; better-performing states can expand.

## Missegregation can also produce nonviable daughters



Stress-associated death and division-linked daughter loss are distinct sources of cell loss.

## A Death-hazard-linked CIN and ploidy selection create a population-composition feedback



A constant-probability WGD branch supplies an additional route into high-ploidy states.